Tobias Hullette

Raleigh, NC
tobias.hullette@gmail.com
(828) 514-2894
linkedin.com/in/tobias-hullette-0087a718a/

EDUCATION

North Carolina State University **August 2024 – May 2026** ***(Anticipated)***
Master of Science | **Aerospace Engineering** | Park Scholar

North Carolina State University **August 2021 – May 2024**
Bachelor of Science | **Aerospace Engineering** | Park Scholar

EXPERIENCE

REACHR – NASA Research Grant **November 2024 – Present**
P.I. (Principal Investigator)

- Designed, analyzed, built, tested, and flew a multi-mode VTOL-to-Cruise, multi-domain air vehicle for post hurricane surveillance and relief delivery.
- Won NASA USRC grant and received $80,000 to continue R&D.
- Manage a ten-person team of PhD, masters, and undergraduate students in design, analysis (FEA and CFD), manufacturing, and testing of full-scale, mission capable composite Unmanned Aerial Vehicle, HERO 2.
- Leads research in implementing autonomous, multi-mode drone technology in the emergency response industry.

RARE Lab – North Carolina State University (Research) **July 2024 – Present**
Graduate Research Assistant

- Design and build automated experimental setups for force-torque and particle image velocimetry rotorcraft aerodynamics studies.
- Run experiments with Arduino and DEWESoft DAQ and LaVision particle image velocimetry.
- Post-process and analyze data in MATLAB, DaVis11, DeweSoft, and NI.
- Presented research at 2025 Vertical Flight Society – Forum 81 conference.

Northrop Grumman Corporation **Summer 2023 & Summer 2024**
Advanced Manufacturing Systems Engineering - Composites Intern

- Obtained a Department of Defense (DoD) Secret Clearance (Active)
- Designed a COTs-based system to identify composite layup vacuum bag leaks on the factory floor, pushed the solution through testing and implementation approval, saving over $1M annually on a single classified program.
- Designed MATLAB apps to visualize real time heated cure table uniformity for flat panel layups.
- Coded a multi-factor model for oven, heated cure table, and autoclave usage for optimizing batch part production scheduling.
- Coordinated intern tasking to help launch the Northrop Grumman Palmdale Fabrication Laboratory.

ISSRL Lab - North Carolina State University (Research) **October 2021 – April 2023**
Mechanical Design and Fabrication Lead

- Designed, analyzed, fabricated, and assembled parts for the MantaRay Underwater Hydrokinetic Energy Harvesting Vehicle funded by Defense Advanced Research Projects Agency (DARPA) and the DoD.
- Led graduate and PhD researchers in vehicle design, FEA analysis, manufacturing, and assembly tasks.
- Built and maintained bill of materials, coordinated material purchasing with vendors, and managed part inventory.

SKILLS

- 2D/3D Design (CATIA, NX, SolidWorks, OnShape, AutoCAD)
- Computational Analysis (Ansys Structural FEA, Ansys Fluent CFD)
- Programming (MATLAB, Python, C++, G-Code)
- Manufacturing (Metallic, Composites, Assemblies)
- GD&T, Vendor Review for Manufacturability
- Problem Solving and Work Ethic
- Leadership and Teamwork
- Writing and Communication